Workstream Architecture Brief

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Flow task evaluation and contribution infrastructure

Workstream manages project guides, task queues, submission packets, automated checks, reviewer routing, evaluation sprints, revision loops, contribution records, payment status, and reputation signals.

Workstream is how Flow measures, certifies, and coordinates useful human-agent work.

Scope: v0.1 first 30 days, with future adapter context for identity, task contracts, settlement, and reputation.

Executive Summary

Workstream is the operating system for useful work inside Flow. It does not try to own every possible execution environment. Instead, it gives every project a guide, every task a locked policy context, every submission an evidence packet, every review a canonical decision, and every accepted task a contribution record before payment and reputation events.

The first 30 days are focused on proving the internal lifecycle:

```
Project Guide -> Task Queue -> Submission Packet -> Checks -> Review
-> Revision / Acceptance / Rejection -> Contribution Record
-> Payment Record -> Reputation Event
```

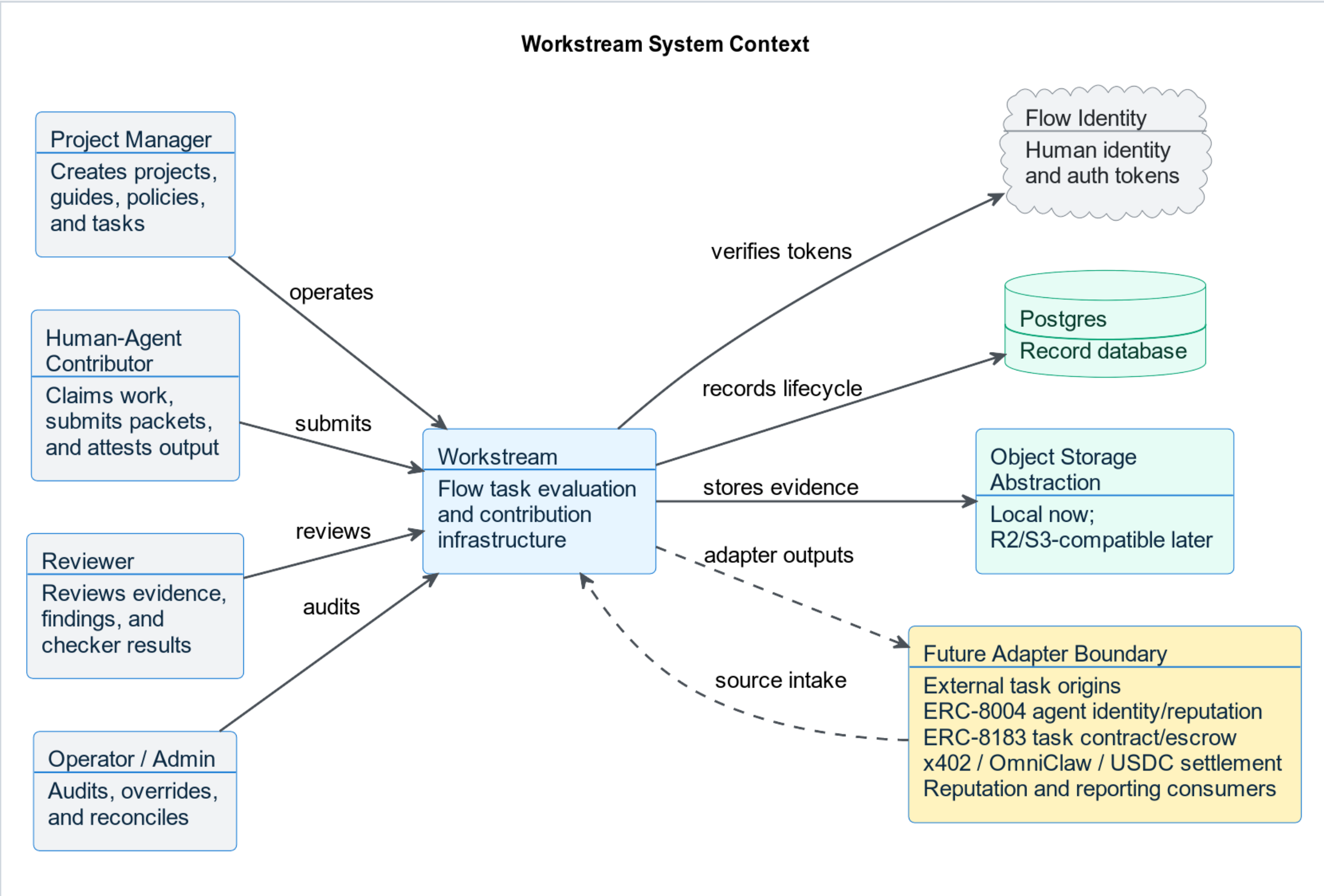
> Current v0.1 is backend-first and internal-loop-first. External source adapters, agent identity writes, task escrow, x402 payment requests, OmniClaw settlement, USDC payouts, public marketplace flows, and automated routing remain adapter boundaries until the internal evaluation loop works with real tasks.

Architecture Principles

Principle	**Meaning**
Source-agnostic, manual-first	v0.1 accepts manual, markdown, or CSV-controlled intake. Future origins normalize into the same task contract.
Flow auth boundary	Workstream verifies Flow-issued tokens. It does not own login, signup, password reset, password storage, or primary sessions.
Modular monolith	FastAPI remains one deployable backend while keeping routers, services, repositories, ports, and adapters separate.
Postgres record database	Local, CI, and production-like development use Postgres as the record database.
Object-storage abstraction	Local filesystem storage is allowed only behind an R2/S3-compatible storage interface.
Async-first execution	Long-running checker work does not block request/ response paths.
Contribution before payment	Accepted work creates a durable contribution record before payment status or reputation events.

C1: System Context

The context diagram shows Workstream as one system inside the broader Flow ecosystem. Workstream owns evaluation and records. It does not own human identity, future agent identity standards, task escrow, settlement rails, or external task origins.

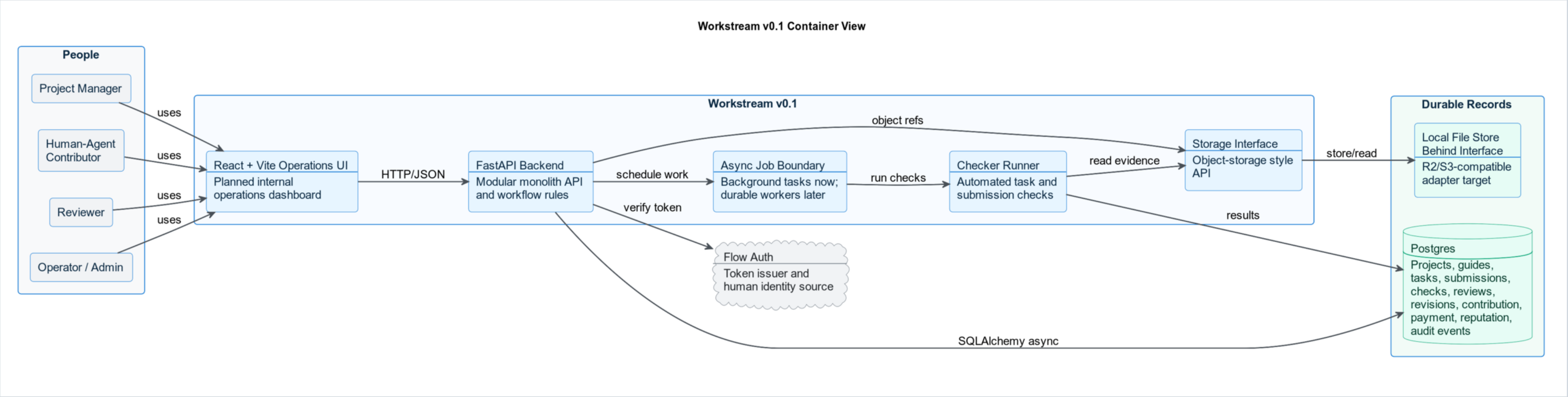


What This Means

- Project managers, contributors, reviewers, and operators interact with Workstream.
- Flow identity remains the human identity and auth source.
- Postgres is the record database.
- Storage sits behind an object-storage abstraction.
- Future origins and protocol rails connect through adapters, not by becoming core Workstream logic.

C2: v0.1 Container View

The container view shows the first 30-day implementation. It is intentionally small: React + Vite for the planned internal operations UI, FastAPI for the backend, Postgres for records, a storage interface for artifacts, and an async checker/job boundary. The Week 1 API demo UI is only a walkthrough client for backend validation; it is not the canonical product frontend.

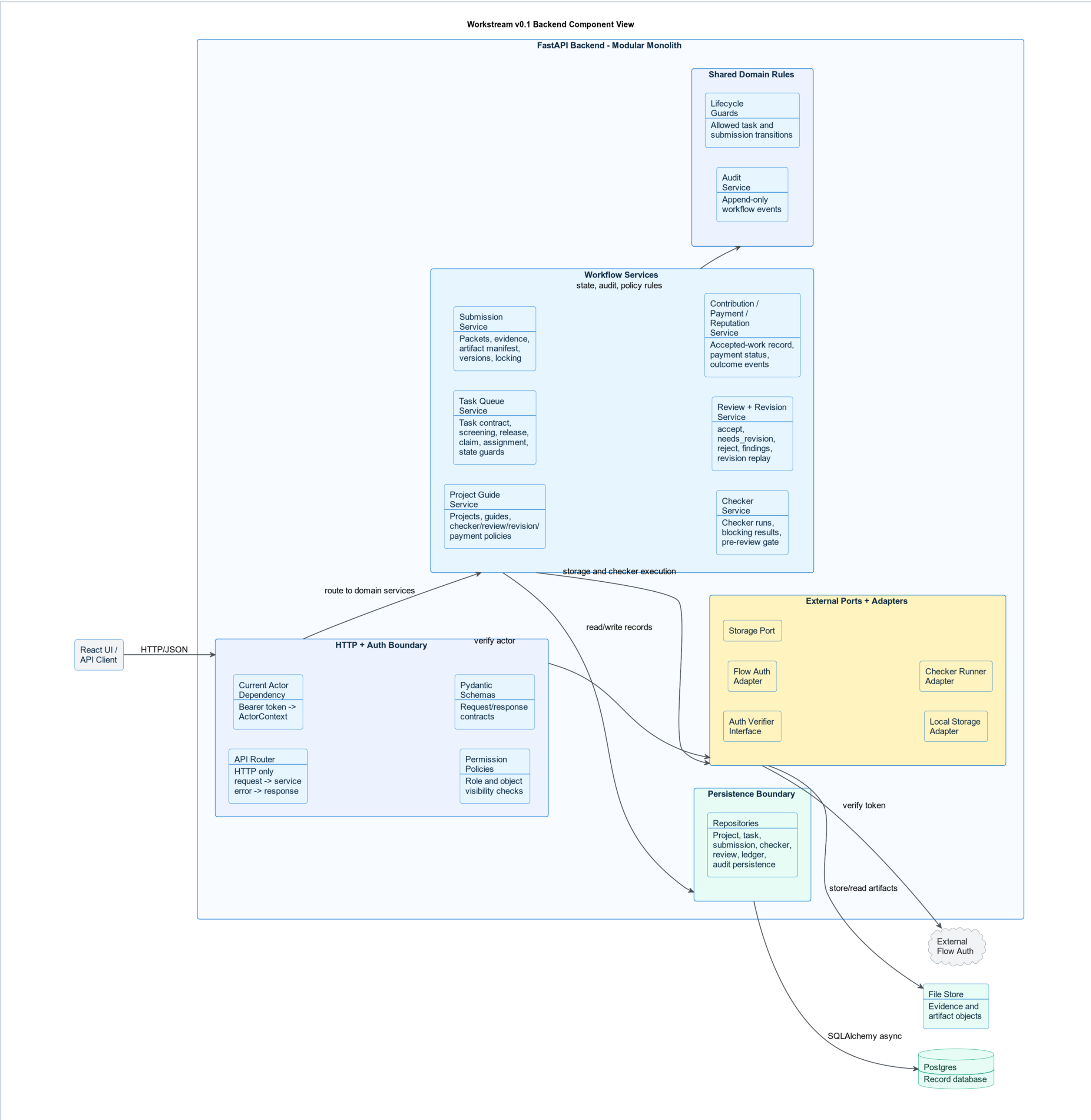


Container Responsibilities

Container	Responsibility
React + Vite operations UI	Planned internal operations dashboard for project, task, submission, review, payment status, and reputation workflows.
FastAPI backend	API contracts, workflow rules, auth dependency, lifecycle guards, module orchestration, and audit writes.
Async job boundary	Non-blocking checker and background work. FastAPI background tasks are acceptable early; durable workers come when retries or distributed execution are needed.
Checker runner	Executes automated checks and stores checker results.
Storage interface	Keeps file/evidence semantics stable while local storage can later move to R2/S3-compatible object storage.
Postgres	Durable record database for the full Workstream lifecycle.

C3: Backend Component View

The backend component view zooms into the FastAPI container. It shows how the modular monolith stays clean without becoming a distributed system too early.

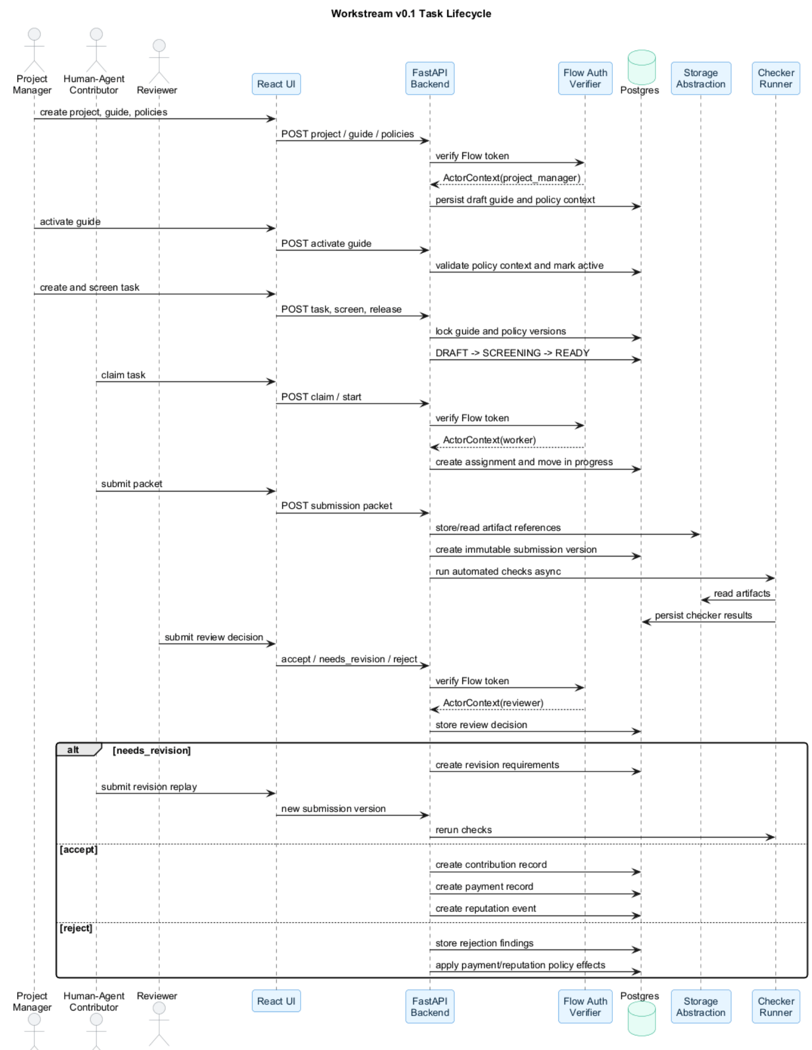


Backend Boundaries

Boundary	Responsibility
HTTP + auth boundary	Routers handle HTTP only. Actor resolution, permission checks, and Pydantic request/response validation stay at the boundary.
Workflow services	Project guide, task queue, submission, checker, review/revision, and contribution/payment/reputation services own business rules.
Shared domain rules	Lifecycle guards and audit writes stay shared instead of being scattered through routers.
Persistence boundary	Repositories own SQLAlchemy async persistence and Postgres access.
External ports/adapters	Flow auth, storage, and checker execution stay behind interfaces.

Lifecycle Sequence

The sequence below shows the narrow v0.1 loop the system must prove before expansion.

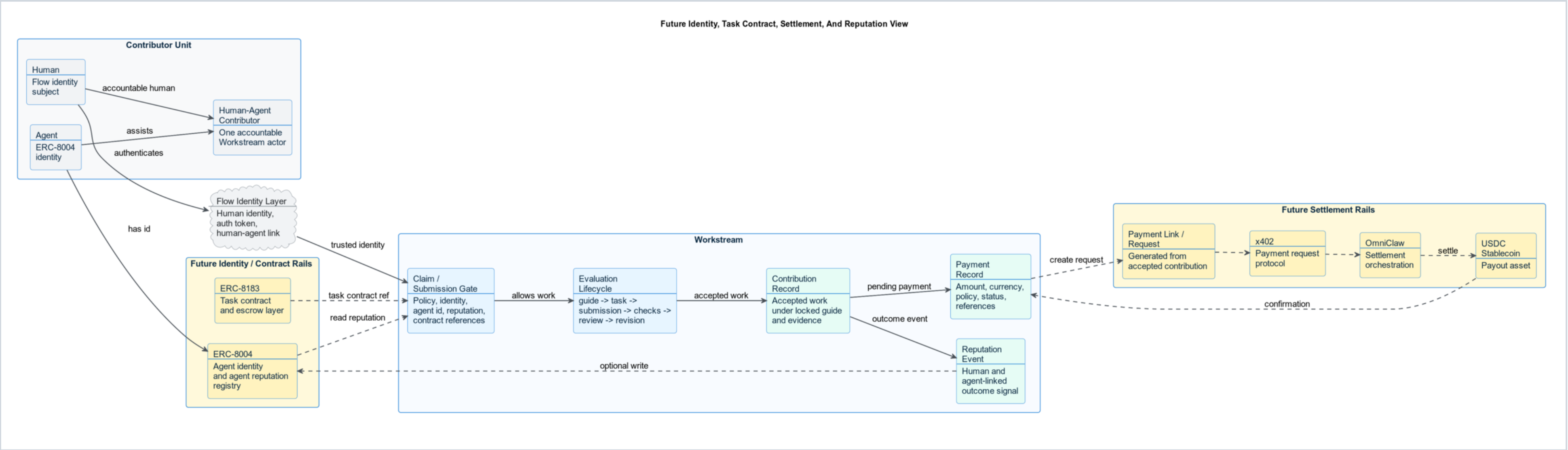


Lifecycle Invariants

- A task cannot enter `READY` without locked guide, checker, review, revision, and payment policy context.
- A worker submission creates a new immutable submission version; locked artifacts are not edited in place.
- Review decisions are exactly `accept`, `needs_revision`, or `reject`.
- `needs_revision` starts a revision loop and must replay prior findings.
- Accepted work creates a contribution record before payment or reputation records.
- Payment status is separate from task acceptance.

Future Identity, Task Contract, Settlement, And Reputation

This view explains the broader architecture direction without moving it into v0.1 scope.



Future Separation Of Concern

Concern	Owner
Human identity and auth	Flow identity layer
Agent identity	ERC-8004
Agent reputation read/write	ERC-8004 through a future Workstream adapter
Task contract and escrow reference	ERC-8183
Evaluation lifecycle	Workstream
Accepted-work certification	Workstream contribution record
Payment policy and payment status	Workstream payment record
Payment request and settlement execution	x402, OmniClaw, and USDC settlement rails

> Future ERC-8004, ERC-8183, x402, OmniClaw, and USDC integrations do not replace Workstream. They use Workstream records. Workstream remains the evaluation, acceptance, contribution, payment-status, and reputation-signal system.

Scope Boundary

Current v0.1 / First 30 Days

- project guide and versioned policy context
- task queue and task records
- assignment and claim flow
- submission packets and evidence
- checker framework and pre-review gate
- human review and revision replay
- contribution records
- payment status records
- reputation events
- audit events

Later Adapter Boundaries

- external origin onboarding and source adapters
- automated routing
- owner-agent execution workspace
- ERC-8004 agent identity and reputation writes
- ERC-8183 task contract and escrow settlement
- x402 payment requests
- OmniClaw settlement orchestration
- USDC payout execution
- marketplace discovery

Closing

Workstream v0.1 succeeds when it can run real internal work from project guide to accepted contribution with evidence, checks, human review, revision discipline, payment status, and reputation events.

The system should expand only after that loop is proven.